Log Monitoring & Analysis – Comprehensive Report

Cyber Security Internship – Task 12

Operating System: Kali Linux

Methodology: Manual Log Monitoring and Analysis

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Abstract

This report presents a comprehensive log monitoring and analysis performed on a Kali Linux system. Multiple system logs including authentication logs, system startup logs, and package management logs were examined. The objective of this task was to understand how logs help detect unauthorized access, monitor system behavior, and support incident detection and forensic investigations.

1. Introduction to Log Monitoring

Log monitoring is a critical security practice used to track system activity, detect suspicious behavior, and investigate security incidents. Every operating system maintains logs that record user activity, system events, and software changes. These logs help security professionals identify threats and maintain system integrity.

2. Log Directory Exploration (/var/log)

The /var/log directory was explored to identify available log files. This directory contains important logs such as boot.log, dpkg.log, btmp, wtmp, apache2 logs, postgresql logs, and system service logs. These logs provide valuable information about system activity, authentication events, and application behavior.

3. System Startup Log Analysis (boot.log)

The boot.log file records system startup events including service initialization, filesystem mounting, network activation, and firewall configuration. The analysis confirmed successful system startup with all critical services functioning normally.

4. Package Management Log Analysis (dpkg.log)

The dpkg.log file records software installation, upgrade, and configuration events. The logs showed installation and configuration of packages such as OpenJDK, Neo4j, and system utilities. This confirms that all software changes are recorded and can be reviewed for security purposes.

5. Authentication Log Analysis (wtmp)

Authentication logs were analyzed using the 'last' command, which reads data from the wtmp log file. The logs showed multiple successful login sessions by the authorized user 'nihal'. This confirms normal user activity and no unauthorized login attempts were observed during analysis.

6. Failed Login Detection (btmp)

Failed login attempts are recorded in the btmp log file. Monitoring failed login attempts helps detect brute-force attacks and unauthorized access attempts. Security teams regularly monitor this log to identify potential threats.

7. Importance of Log Monitoring in Cyber Security

1. Detect unauthorized access attempts
2. Monitor system and user activity
3. Detect malware and intrusion attempts
4. Support forensic investigations
5. Help maintain system security and compliance

8. SIEM (Security Information and Event Management)

SIEM systems collect, analyze, and correlate logs from multiple systems. Popular SIEM tools include Splunk, ELK Stack, and IBM QRadar. These tools help detect security incidents in real time and generate alerts for security teams.

9. Incident Detection and Security Monitoring

Log monitoring is essential for detecting security incidents such as unauthorized access, malware activity, and system compromise. By analyzing logs, security professionals can identify threats early and respond quickly.

10. Conclusion

This task provided hands-on experience in monitoring and analyzing Linux system logs. The analysis confirmed normal system operation and demonstrated the importance of log monitoring in detecting and preventing security threats. Log analysis is a fundamental skill for cyber security professionals.